The Hurewicz Property on the Vietoris Hyperspace of Compact Subsets

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Definitions and Background

Topological Operators

Generally speaking, we'll refer to any class function defined on topological spaces to be a *topological operator*. As usual, we'll tend to collapse the notation $(X, \mathscr{T})$ to just X; when other topologies on a set are considered, we will employ notation to distinguish them.

The topological operators we'll be concerned with here are the following, relative to a given space X.

- $\mathscr{T}$: $\mathscr{T}_X$ is the collection of open subsets U of X so that $U \neq X$ and $U \neq \emptyset$.
- $\mathcal{O}$: $\mathcal{O}_X$ is the collection of all $\mathscr{U} \subseteq \mathscr{T}_X$ so that $X = \bigcup \mathscr{U}$; that is, the set of non-trivial open covers of X.
- Ω: Ω_X is the set of all ω-covers of X; that is, the set of $\mathscr{U} \in \mathcal{O}_X$ so that every finite subset of X is contained in a member of $\mathscr{U}$.
- $\mathcal{K}$: $\mathcal{K}_X$ is the set of all k-covers of X; that is, the set of $\mathscr{U} \in \mathcal{O}_X$ so that every compact subset of X is contained in a member of $\mathscr{U}$.

For any space X,

$$\mathcal{K}_X \subseteq \Omega_X \subseteq \mathcal{O}_X,$$

and they are, in general, strict subsets.

- Note that any cover of $\mathbb{R}$ by balls of a fixed radius is an open cover which is not an ω-cover.
- Note that the set of all open subsets of $\mathbb{R}$ with a fixed bound on their Lebesgue measure is an ω-cover which is not a k-cover.

Note that, since ω- and k-covers are non-trivial, they are all *large* covers; that is, every finite (resp. compact) set is contained in infinitely many members of a given ω-cover (resp. k-cover).

Hence, if $\mathscr{U}$ is an ω-cover (resp. k-cover), then so is $\mathscr{U} \setminus \mathscr{F}$ for any $\mathscr{F} \in [\mathscr{U}]^{<\omega}$.

We use $K(X)$ to denote the set of non-empty compact subsets of X.

The Vietoris Topology

The Vietoris topology on the set $K(X)$ has as its basis sets of the form

$$[U_1, U_2, \ldots, U_n] = \left\{ K \in K(X) : K \subseteq \bigcup_{j=1}^{n} U_n \wedge (\forall j)\ K \cap U_j \neq \emptyset \right\}.$$

We use $\mathbb{K}(X)$ to denote the set $K(X)$ endowed with the Vietoris topology and we use $\mathcal{P}_{\text{fin}}(X)$ to denote the set $[X]^{<\omega}$ with the topology it inherits as a subspace of $\mathbb{K}(X)$.

Selection Principles

For collections/topological operators $\mathcal{A}$ and $\mathcal{B}$,

$$\mathsf{S}_{\mathrm{fin}}(\mathcal{A},\mathcal{B}) \equiv (\forall \mathscr{U} \in \mathcal{A}^{\omega}) \left(\exists \mathscr{F} \in \prod_{n\in\omega} [\mathscr{U}_n]^{<\omega} \right) \bigcup_{n\in\omega} \mathscr{F}_n \in \mathcal{B}$$

and

$$\mathsf{S}_{1}(\mathcal{A},\mathcal{B}) \equiv (\forall \mathscr{U} \in \mathcal{A}^{\omega}) \left(\exists U \in \prod_{n\in\omega} \mathscr{U}_n \right) \{U_n : n \in \omega\} \in \mathcal{B}$$

A space X is said to be

- *Menger* (resp. *Rothberger*) if it satisfies $\mathsf{S}_{\mathrm{fin}}(\mathcal{O},\mathcal{O})$ (resp. $\mathsf{S}_1(\mathcal{O},\mathcal{O})$).
- *ω-Menger* (resp. *ω-Rothberger*) if it satisfies $\mathsf{S}_{\mathrm{fin}}(\Omega,\Omega)$ (resp. $\mathsf{S}_1(\Omega,\Omega)$).
- *k-Menger* (resp. *k-Rothberger*) if it satisfies $\mathsf{S}_{\mathrm{fin}}(\mathcal{K},\mathcal{K})$ (resp. $\mathsf{S}_1(\mathcal{K},\mathcal{K})$).

The Hurewicz Property

A space X has the *Hurewicz property* (or is said to be *Hurewicz*) if

$$(\forall \mathscr{U} \in \mathcal{O}_X^\omega)\left(\exists \mathscr{F} \in \prod_{n\in\omega} [\mathscr{U}_n]^{<\omega}\right)\ X \subseteq \bigcup_{m\in\omega} \bigcap_{n\geqslant m} \bigcup \mathscr{F}_n.$$

- Every σ-compact space is Hurewicz.
- Every Hurewicz space is Menger.

For each of the selection principles and the Hurewicz property, there are corresponding games with ω rounds for two players, One and Two.

- $\mathsf{G}_{\mathrm{fin}}(\mathcal{A}, \mathcal{B})$ denotes the finite-selection game corresponding to $\mathsf{S}_{\mathrm{fin}}(\mathcal{A}, \mathcal{B})$.
- $\mathsf{G}_1(\mathcal{A}, \mathcal{B})$ denotes the single-selection game corresponding to $\mathsf{S}_1(\mathcal{A}, \mathcal{B})$.
- $\mathsf{H}(X)$ denotes the Hurewicz game on X.

We will use the notation

- $\mathrm{I} \uparrow \mathcal{G}$ to mean that One has a winning strategy in the game $\mathcal{G}$.
- $\mathrm{I} \underset{\text{pre}}{\uparrow} \mathcal{G}$ to mean that One has a pre-determined winning strategy in $\mathcal{G}$; that is, a winning strategy that only depends on the turn number, and not Two's responses.

In general,

$$\mathsf{S}_{\square}(\mathcal{A}, \mathcal{B}) \iff \mathrm{I} \underset{\text{pre}}{\not\uparrow} \mathsf{G}_{\square}(\mathcal{A}, \mathcal{B}).$$

- (Hurewicz 1925)

$$\mathrm{I} \uparrow \mathsf{G}_{\mathrm{fin}}(\mathcal{O}_X, \mathcal{O}_X) \iff \mathrm{I} \underset{\mathrm{pre}}{\uparrow} \mathsf{G}_{\mathrm{fin}}(\mathcal{O}_X, \mathcal{O}_X)$$

- (Pawlikowski 1994)

$$\mathrm{I} \uparrow \mathsf{G}_1(\mathcal{O}_X, \mathcal{O}_X) \iff \mathrm{I} \underset{\mathrm{pre}}{\uparrow} \mathsf{G}_1(\mathcal{O}_X, \mathcal{O}_X)$$

- (Scheepers 1997)

$$\mathrm{I} \uparrow \mathsf{G}_{\square}(\Omega_X, \Omega_X) \iff \mathrm{I} \underset{\mathrm{pre}}{\uparrow} \mathsf{G}_{\square}(\Omega_X, \Omega_X)$$

- (C. & Holshouser 2022)

$$\mathrm{I} \uparrow \mathsf{G}_{\square}(\mathcal{K}_X, \mathcal{K}_X) \iff \mathrm{I} \underset{\mathrm{pre}}{\uparrow} \mathsf{G}_{\square}(\mathcal{K}_X, \mathcal{K}_X)$$

- (Scheepers 1996)

$$\mathrm{I} \uparrow \mathsf{H}(X) \iff \mathrm{I} \underset{\mathrm{pre}}{\uparrow} \mathsf{H}(X)$$

Inspiration

Groupability (Kočinac & Scheepers 2003)

For a class $\mathfrak{C}$, a member $\mathscr{C} \in \mathfrak{C}$ is *groupable* if there is a function $\phi : \mathscr{C} \to \omega$ so that ϕ is finite-to-one and, for any infinite $J \subseteq \omega$, $\bigcup\{\phi^{-1}(j) : j \in J\} \in \mathfrak{C}$.

For a topological operator $\mathfrak{O}$, we will let $\mathfrak{O}^{gp}$ be the operator that produces the groupable elements of $\mathfrak{O}_X$.

Theorem (Kočinac & Scheepers 2003)

For a space X, every finite power of X is Hurewicz if and only if $X \models \mathsf{S}_{\mathrm{fin}}(\Omega, \Omega^{gp})$.

Theorem (Just, Miller, & Scheepers 1996)

For a space X, every finite power of X is Menger if and only if X is ω-Menger.

Theorem (Sakai 1988)

For a space X, every finite power of X is Rothberger if and only if X is ω-Rothberger.

Theorem (C. & Holshouser 2022)

For a space X,

- $\mathcal{P}_{\text{fin}}(X)$ is Menger if and only if X is ω-Menger.
- $\mathcal{P}_{\text{fin}}(X)$ is Rothberger if and only if X is ω-Rothberger.

Theorem (C. & Holshouser 2022)

For a space X, $\mathbb{K}(X)$ is Menger if and only if X is k-Menger.

There is not an analogous result characterizing when $\mathbb{K}(X)$ is Rothberger since $\mathbb{R}$ is an example of a space which is k-Rothberger but $\mathbb{K}(\mathbb{R})$ is not Rothberger.

The Hyperspace Correspondence

Some useful facts:

- If $K_0 \subseteq X$ is compact, then $\{K \in \mathbb{K}(X) : K \subseteq K_0\}$ is compact in $\mathbb{K}(X)$.
- If $\mathbf{K} \subseteq \mathbb{K}(X)$ is compact, then $\bigcup \mathbf{K} \subseteq X$ is compact.
- If $\mathscr{U}$ is a k-cover of X, then $\{[U] : U \in \mathscr{U}\}$ is a k-cover of $\mathbb{K}(X)$.
- (Analogous to The Wallace Theorem) If $\mathscr{U}$ is a k-cover of $\mathbb{K}(X)$, then

$$\{V \in \mathscr{T}_X : (\exists U \in \mathscr{U})\ [V] \subseteq U\}$$

is a k-cover of X.

Note that not every open cover of $\mathbb{K}(X)$ has the property described above. Consider

$$\mathscr{U} := \{[U_1, U_2, \ldots, U_n] : (\forall j)\ \mathrm{diam}(U_j) < \varepsilon\}$$

for your favorite $\varepsilon > 0$. $\mathscr{U}$ is an open cover of $\mathbb{K}(X)$, but note that the set $K := [-2\varepsilon, 2\varepsilon]$ is a compact subset of $\mathbb{R}$ and there is no open $V \subseteq \mathbb{R}$ so that

$$K \in [V] \subseteq [U_1, U_2, \ldots, U_n]$$

for some set of U_j as specified above (which can be verified by picking out a well-chosen singleton element from K).

Lemma

For any sequence $\mathscr{U} \in \left(\mathcal{K}_X^{gp}\right)^\omega$, there exists $\mathscr{V} \in \left(\mathcal{K}_X^{gp}\right)^\omega$ so that $\{\mathscr{V}_n : n \in \omega\}$ is pairwise disjoint and $\mathscr{V}_n \subseteq \mathscr{U}_n$ for each $n \in \omega$.

Lemma

Suppose $X \models \mathsf{S}_{\mathrm{fin}}(\mathcal{K}_X, \mathcal{K}_X^{gp})$. Then for any $\mathscr{U} \in \mathcal{K}_X^\omega$, there exists $\mathscr{F} \in \prod_{n\in\omega} [\mathscr{U}_n]^{<\omega}$ so that $\{\mathscr{F}_n : n \in \omega\}$ is pairwise disjoint and

$$(\forall K \in K(X))(\exists m \in \omega)(\forall n \geqslant m)(\exists V \in \mathscr{F}_n)\ K \subseteq V.$$

Theorem (C. 2023)

For a space X, the following are equivalent.

- $\mathbb{K}(X)$ is Hurewicz.
- $\mathbb{K}(X) \models \mathsf{S}_{\mathrm{fin}}(\mathcal{K}, \mathcal{K}^{gp})$.
- $X \models \mathsf{S}_{\mathrm{fin}}(\mathcal{K}, \mathcal{K}^{gp})$.
- X is k-Menger and every countable k-cover of X is groupable.
- $\mathrm{I} \not\models \mathsf{G}_{\mathrm{fin}}(\mathcal{K}_X, \mathcal{K}_X^{gp})$.

$\mathbb{K}(X)$ is Hurewicz $\implies X \models \mathsf{S}_{\text{fin}}(\mathcal{K}, \mathcal{K}^{gp})$

First note that $\mathbb{K}(X)$ is Menger which implies that X is k-Menger. So we need only show each countable k-cover of X is groupable.

Given $\mathscr{U} = \{U_n : n \in \omega\} \in \mathcal{K}_X$, we define a strategy σ for One in $\mathsf{H}(\mathbb{K}(X))$.

$$\sigma(\langle\rangle): \qquad [U_0] \quad [U_1] \quad [U_2] \quad [U_3] \quad [U_4] \quad [U_5] \quad [U_6] \quad \cdots$$

$$\sigma(\langle\rangle): \qquad [U_0] \quad [U_1] \quad [U_2] \quad [U_3] \quad [U_4] \quad [U_5] \quad [U_6] \quad \cdots$$

$$\begin{array}{cccccccccc} & & & & [U_2] & & [U_4] & [U_5] & \longrightarrow & \mathscr{F}_0 \\ \sigma(\langle\rangle): & [U_0] & [U_1] & & [U_2] & [U_3] & [U_4] & [U_5] & [U_6] & \cdots \end{array}$$

$\sigma(\langle\rangle):$	$[U_0]$	$[U_1]$	$[U_2]$	$[U_3]$	$[U_4]$	$[U_5]$	$[U_6]$	$\cdots$
$\sigma(\langle\mathscr{F}_0\rangle):$	$[U_6]$	$[U_7]$	$[U_8]$	$[U_9]$	$[U_{10}]$	$[U_{11}]$	$[U_{12}]$	$\cdots$

$\sigma(\langle\rangle):$	$[U_0]$	$[U_1]$	$[U_2]$	$[U_3]$	$[U_4]$	$[U_5]$	$[U_6]$	$\cdots$
$\sigma(\langle\mathscr{F}_0\rangle):$	$[U_6]$	$[U_7]$	$[U_8]$	$[U_9]$	$[U_{10}]$	$[U_{11}]$	$[U_{12}]$	$\cdots$

$\sigma(\langle\rangle):$	$[U_0]$	$[U_1]$	$[U_2]$	$[U_3]$	$[U_4]$	$[U_5]$	$[U_6]$	$\cdots$
$\sigma(\langle\mathscr{F}_0\rangle):$	$[U_6]$	$[U_7]$	$[U_8]$	$[U_9]$	$[U_{10}]$	$[U_{11}]$	$[U_{12}]$	$\cdots$

$$
\begin{array}{lcccccccc}
\sigma(\langle\rangle): & [U_0] & [U_1] & [U_2] & [U_3] & [U_4] & [U_5] & [U_6] & \cdots \\
\sigma(\langle\mathscr{F}_0\rangle): & [U_6] & [U_7] & [U_8] & [U_9] & [U_{10}] & [U_{11}] & [U_{12}] & \cdots \\
\sigma(\cdots): & [U_{10}] & [U_{11}] & [U_{12}] & [U_{13}] & [U_{14}] & [U_{15}] & [U_{16}] & \cdots \\
 & & & & \vdots & & & &
\end{array}
$$

This strategy cannot be winning, so Two can make selections of finite sets of integers F_n so that $\max F_n < \min F_{n+1}$ and the corresponding $\{[U_j] : j \in F_n\}$ satisfy the covering criterion on $\mathbb{K}(X)$.

From here, it can be shown that partitioning $\{U_n : n \in \omega\}$ according to the $\max F_n$ establishes groupability.

$X \models \mathsf{S}_{\mathrm{fin}}(\mathcal{K}, \mathcal{K}^{gp}) \implies \mathbb{K}(X)$ is Hurewicz

Let $\langle \mathscr{U}_n : n \in \omega \rangle$ be a sequence of open covers of $\mathbb{K}(X)$ and let $\mathscr{U}_n^*$ denote the closure of $\mathscr{U}_n$ under finite unions. We fix $\gamma_n : \mathscr{U}_n^* \to [\mathscr{U}_n]^{<\omega}$ to be so that $U = \bigcup \gamma_n(U)$.

Now,

$$\mathscr{V}_n = \{V \in \mathscr{T}_X : (\exists U \in \mathscr{U}_n^*)\ [V] \subseteq U\}$$

is a k-cover of X. We fix $\rho_n : \mathscr{V}_n \to \mathscr{U}_n^*$ to be so that $[V] \subseteq \rho_n(V)$.

By a previous lemma, there exists $\mathscr{G} \in \prod_{n\in\omega} [\mathscr{V}_n]^{<\omega}$ so that $\{\mathscr{G}_n : n \in \omega\}$ is pairwise disjoint and

$$(\forall K \in K(X))(\exists m \in \omega)(\forall n \geqslant m)(\exists V \in \mathscr{G}_n)\ K \subseteq V.$$

Then, defining $\mathscr{F}_n = \bigcup\{\gamma_n \circ \rho_n(V) : V \in \mathscr{G}_n\}$ gives a finite selection from $\mathscr{U}_n$ so that $\langle \mathscr{F}_n : n \in \omega \rangle$ satisfies the Hurewicz covering criterion.

$\mathbb{K}(X)$ is Hurewicz $\implies \mathbb{K}(X) \models \mathsf{S}_{\mathrm{fin}}(\mathcal{K}, \mathcal{K}^{gp})$

From above, we have that $X \models \mathsf{S}_{\mathrm{fin}}(\mathcal{K}, \mathcal{K}^{gp})$. Hence, $\mathbb{K}(X)$ is k-Menger, so we need only show that each countable k-cover of $\mathbb{K}(X)$ is groupable. This is done in a similar way as above by defining a strategy for One in $\mathsf{H}(\mathbb{K}(X))$, but by passing to adequate countable refinements relative to X.

Function Spaces

$C_k(X)$ denotes the set $C(X)$ of continuous real-valued functions with the topology of uniform convergence on compacta. It has a basis sets of the form

$$[f; K, \varepsilon] = \{g \in C_k(X) : \sup\{|f(x) - g(x)| : x \in K\} < \varepsilon\}$$

where $f \in C_k(X)$, $K \in K(X)$, and $\varepsilon > 0$.

For a space X and $x \in X$,

$$\Omega_{X,x} = \{A \subseteq X : x \in \mathrm{cl}(A) \setminus A\}.$$

Then X has

- *countable fan-tightness* at x if $X \models \mathsf{S}_{\mathrm{fin}}(\Omega_{X,x}, \Omega_{X,x})$.
- *countable strong fan-tightness* at x if $X \models \mathsf{S}_1(\Omega_{X,x}, \Omega_{X,x})$.

A space X has the *Reznichenko property* at x if every countable element of $\Omega_{X,x}$ is groupable.

Theorem (Kočinac 2003)

For a Tychonoff space X,

$$\mathrm{I} \not\models \mathsf{G}_{\square}(\mathcal{K}_X, \mathcal{K}_X^{gp}) \implies C_k(X) \models \mathsf{S}_{\square}(\Omega_{C_k(X),\mathbf{0}}, \Omega^{gp}_{C_k(X),\mathbf{0}}).$$

Question (Kočinac 2003)

Do the implications above reverse?

Answer: Yes.

Theorem (C. 2023)

For any Tychonoff space, the following are equivalent.

- $X \models \mathsf{S}_{\square}(\mathcal{K}, \mathcal{K}^{gp})$
- $\mathrm{I} \not\models \mathsf{G}_{\square}(\mathcal{K}, \mathcal{K}^{gp})$
- $C_k(X)$ is Reznichenko and has countable (strong) fan-tightness.
- $C_k(X) \models \mathsf{S}_{\square}(\Omega_{C_k(X),\mathbf{0}}, \Omega^{gp}_{C_k(X),\mathbf{0}})$.

We'll fix some translations.

- $\mathfrak{u} : C(X) \times \omega \to \mathscr{T}_X \cup \{X\}$ defined by $f^{-1}((-2^{-n}, 2^{-n}))$.
- $\mathbb{U} : \Omega_{C_k(X),\mathbf{0}} \times \omega \to \mathcal{K}_X$ defined by

$$\begin{aligned}&\mathbb{U}(\mathscr{F}, n)\\ &= \{U \in \mathscr{T}_X : (\exists K \in \mathbb{K}(X))(\exists f \in \mathscr{F})\ K \subseteq U \subseteq \mathfrak{u}(f, n)\}.\end{aligned}$$

- $\mathscr{D} : \mathcal{K}_X \to \Omega_{C_k(X),\mathbf{0}}$ defined by

$$\mathscr{D}(\mathscr{U}) = \{f \in C(X) : (\exists U \in \mathscr{U})\ X \setminus U \subseteq f^{-1}(1)\}.$$

$C_k(X) \models \mathsf{S}_{\mathrm{fin}}(\Omega_{C_k(X),\mathbf{0}}, \Omega^{gp}_{C_k(X),\mathbf{0}}) \implies X \models \mathsf{S}_{\mathrm{fin}}(\mathcal{K}, \mathcal{K}^{gp})$

First,

$$C_k(X) \models \mathsf{S}_{\mathrm{fin}}(\Omega_{C_k(X),\mathbf{0}}, \Omega_{C_k(X),\mathbf{0}}) \implies X \models \mathsf{S}_{\mathrm{fin}}(\mathcal{K}, \mathcal{K}).$$

Let $\langle \mathscr{U}_n : n \in \omega \rangle$ be a sequence of k-covers of X. Then $\langle \mathscr{D}(\mathscr{U}_n) : n \in \omega \rangle$ is a sequence of $\Omega_{C_k(X),\mathbf{0}}$ so we can choose $\mathscr{G}_n \in [\mathscr{D}(\mathscr{U}_n)]^{<\omega}$ so that $\bigcup_{n\in\omega} \mathscr{G}_n \in \Omega_{C_k(X),\mathbf{0}}$.

We can let $\mathscr{F}_n \in [\mathscr{U}_n]^{<\omega}$ be so that, for each $f \in \mathscr{G}_n$, there is some $U \in \mathscr{F}_n$ so that $X \setminus U \subseteq f^{-1}(1)$.

To see that $\bigcup_{n\in\omega} \mathscr{F}_n \in \mathcal{K}_X$, given any $K \in \mathbb{K}(X)$, consider the neighborhood $[\mathbf{0}; K, 1/2]$. There exist $n \in \omega$ and $f \in \mathscr{G}_n$ so that $f \in [\mathbf{0}; K, 1/2]$, from which it follows that K is covered by a member of $\mathscr{F}_n$.

Now we show every countable k-cover of X is groupable by letting $\mathscr{U} = \{U_n : n \in \omega\} \in \mathcal{K}_X$ and defining a strategy σ for One in $\mathsf{G}_{\mathrm{fin}}(\mathcal{K}_X, \mathcal{K}_X)$.

To get off the ground, for $\langle K, U\rangle \in \mathbb{K}(X) \times \mathscr{U}$ so that $K \subseteq U$, let $f_{K,U} \in C(X)$ be so that $f_{K,U}[K] = \{0\}$ and $f_{K,U}[X \setminus U] = \{1\}$. Then

$$\mathscr{F}_0 := \{f_{K,U} : K \in \mathbb{K}(X), U \in \mathscr{U}, K \subseteq U\} \in \Omega_{C_k(X),\mathbf{0}}.$$

Apply $\mathsf{S}_{\mathrm{fin}}(\Omega_{C_k(X),\mathbf{0}}, \Omega^{gp}_{C_k(X),\mathbf{0}})$ to $\mathscr{F}_0$ to obtain

$$\begin{array}{cccc} g_0 & g_1 & \cdots & g_{M_0} \\ g_{M_0+1} & g_{M_0+2} & \cdots & g_{M_1} \\ \vdots & & & \end{array}$$

where $\{g_n : n \in \omega\} \subseteq \mathscr{F}_0$ and the partition ψ defined by $\psi(g_n) = \min\{k \in \omega : n \leqslant M_k\}$ satisfies the groupability criterion.

In particular, for $K \in \mathbb{K}(X)$, there exists $m \in \omega$ so that, for every $n \geqslant m+1$,

$$[\mathbf{0}; K, 1/2] \cap \{g_\ell : M_{n-1} < \ell \leqslant M_n\} \neq \emptyset.$$

Now define $\gamma : \omega \to \omega$ by

$$\gamma(n) = \min\{k \in \omega : g_n[X \setminus U_k] = \{1\}\}$$

and

$$\sigma(\langle\rangle) = \{U_{\gamma(n)} : n \in \omega\} = \mathscr{W}.$$

Note that $\sigma(\langle\rangle) \in \mathcal{K}_X$.

We will associate Two's moves with finite selections from the g_n, perhaps via some fixed choice function, or perhaps via

$$W \mapsto \min\{k \in \omega : U_{\gamma(k)} = W\}, \mathscr{W} \to \omega.$$

Consider $\mathscr{F}_0 \in [\sigma(\langle\rangle)]^{<\omega}$ and $\Lambda_0 \in [\omega]^{<\omega}$ so that $\mathscr{F}_0 = \{U_{\gamma(n)} : n \in \Lambda_0\}$. Then we can define

$$\mu_0 = \max\{\psi(g_n) : n \in \Lambda_0\}$$

and

$$\nu_0 = \max\{\gamma(n) : n \leqslant M_{\mu_0}\}.$$

The μ_0 segments off entire "blocks" of the g_n according to the partition and the ν_0 "chops off" the $U_{\gamma(n)}$ that are mapped by the initial segment corresponding to μ_0.

Two's first selection

$\mathscr{F}_0$

$U_{\gamma(k_0)}$

$U_{\gamma(k_1)}$

$U_{\gamma(k_2)}$

$U_{\gamma(k_3)}$

$U_{\gamma(k_4)}$

g_0
g_1
g_2
g_3
$\vdots$
g_{M_0}

g_{M_0+1}
g_{M_0+2}
g_{M_0+3}
$\vdots$
g_{M_1}

g_{M_1+1}
$\vdots$
g_{M_2}

g_{M_2+1}
$\vdots$

Two's first selection

$\mathscr{F}_0$

$U_{\gamma(k_0)}$

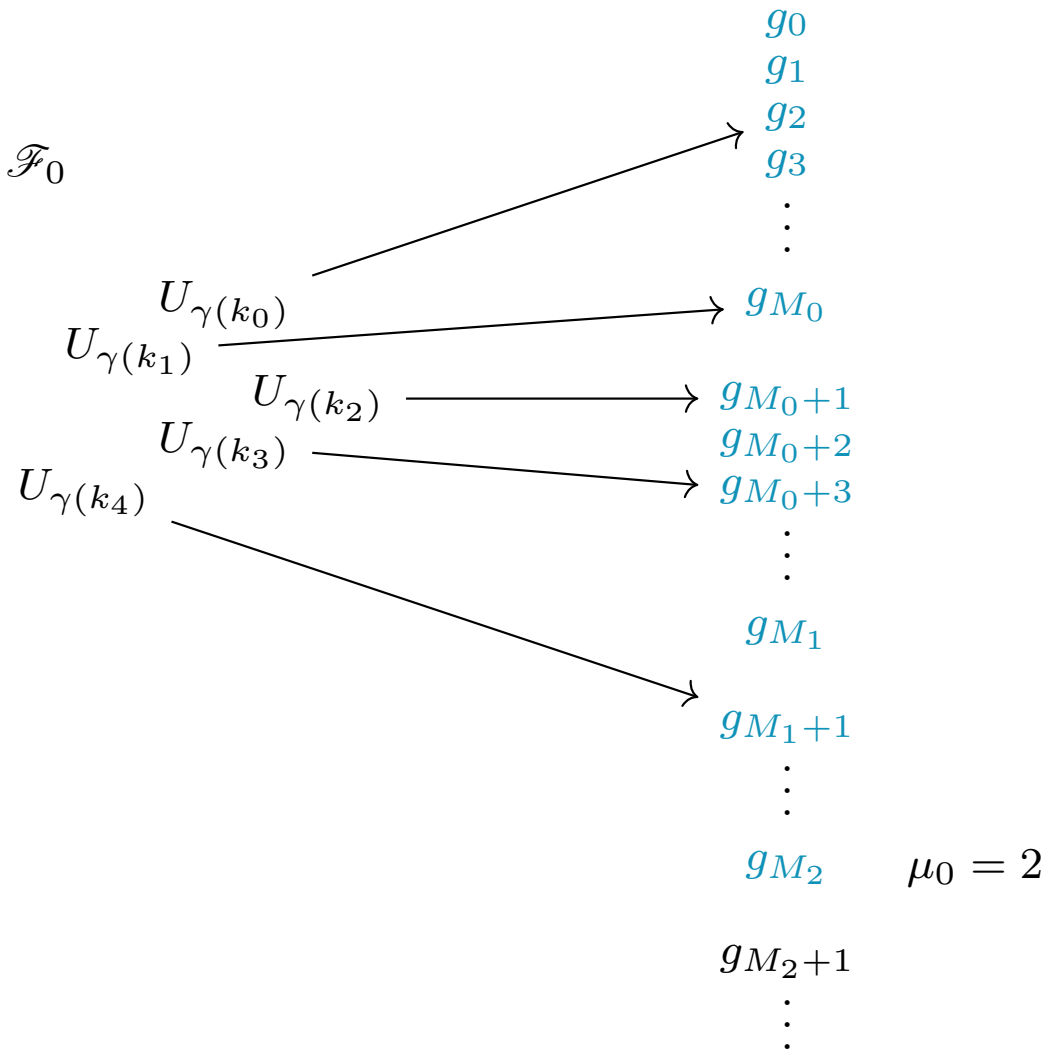
$U_{\gamma(k_1)}$

$U_{\gamma(k_2)}$

$U_{\gamma(k_3)}$

$U_{\gamma(k_4)}$

g_0
g_1
g_2
g_3
$\vdots$
g_{M_0}
g_{M_0+1}
g_{M_0+2}
g_{M_0+3}
$\vdots$
g_{M_1}
g_{M_1+1}
$\vdots$
g_{M_2}
g_{M_2+1}
$\vdots$


$\mathscr{F}_0$
$U_{\gamma(k_0)}$
$U_{\gamma(k_1)}$
$U_{\gamma(k_2)}$
$U_{\gamma(k_3)}$
$U_{\gamma(k_4)}$
g_0
g_1
g_2
g_3
g_{M_0}
g_{M_0+1}
g_{M_0+2}
g_{M_0+3}
g_{M_1}
g_{M_1+1}
g_{M_2}
g_{M_2+1}
$\mu_0 = 2$

$$
\begin{array}{ccc}
U_{\gamma(0)} & \longleftarrow & g_0 \\
U_{\gamma(1)} & \longleftarrow & g_1 \\
U_{\gamma(2)} & \longleftarrow & g_2 \\
 & & g_3 \\
 & \vdots & \vdots \\
 & & g_{M_0} \\
 & & g_{M_0+1} \\
 & & g_{M_0+2} \\
\vdots & & g_{M_0+3} \\
 & & \vdots \\
 & & g_{M_1} \\
 & & g_{M_1+1} \\
 & & \vdots \\
U_{\gamma(M_2)} & \longleftarrow & g_{M_2} \qquad \mu_0 = 2 \\
 & & g_{M_2+1} \\
\nu_0 = \max\{\gamma(n) : n \leqslant M_{\mu_0}\} & & \vdots
\end{array}
$$

Then we define

$$\sigma(\langle \mathscr{F}_0 \rangle) = \{U_n : n > \nu_0\} \cap \mathscr{W} \in \mathcal{K}_X.$$

We repeat the same sort of auxiliary associations with the g_n, but this time noting that Two's response here will only be associated to g_n where $n > M_{\mu_0}$.

Two's second selection

$$\begin{array}{lll}
 & & g_{M_2+1} \\
 & & g_{M_2+2} \\
\mathscr{F}_1 & & g_{M_2+3} \\
 & & \vdots \\
 & U_{\gamma(k_5)} & g_{M_3} \\
 & & \\
U_{\gamma(k_6)} & & g_{M_3+1} \\
 & & g_{M_3+2} \\
 & U_{\gamma(k_7)} & g_{M_3+3} \\
 & & \vdots \\
 & & g_{M_4} \\
 & & \\
 & & g_{M_4+1} \\
 & & \vdots \\
 & & g_{M_5} \\
 & & \\
 & & g_{M_5+1} \\
 & & \vdots
\end{array}$$

Two's second selection

$\mathscr{F}_1$

$U_{\gamma(k_5)}$

$U_{\gamma(k_6)}$

$U_{\gamma(k_7)}$

g_{M_2+1}
g_{M_2+2}
g_{M_2+3}
$\vdots$
g_{M_3}

g_{M_3+1}
g_{M_3+2}
g_{M_3+3}
$\vdots$
g_{M_4}

g_{M_4+1}
$\vdots$
g_{M_5}

g_{M_5+1}
$\vdots$

$$
\begin{tikzcd}
\mathscr{F}_1 & & g_{M_2+1} \\
 & & g_{M_2+2} \\
 & & g_{M_2+3} \\
 & & \vdots \\
 & U_{\gamma(k_5)} \arrow[uur] & g_{M_3} \\
 & U_{\gamma(k_6)} \arrow[urr] & g_{M_3+1} \\
 & & g_{M_3+2} \\
 & U_{\gamma(k_7)} \arrow[rr] & g_{M_3+3} \\
 & & \vdots \\
 & & g_{M_4} \qquad \mu_1 = 4 \\
 & & g_{M_4+1} \\
 & & \vdots \\
 & & g_{M_5} \\
 & & g_{M_5+1} \\
 & & \vdots
\end{tikzcd}
$$

$$
\begin{array}{ccc}
U_{\gamma(M_2+1)} & \longleftarrow & g_{M_2+1} \\
U_{\gamma(M_2+2)} & \longleftarrow & g_{M_2+2} \\
U_{\gamma(M_2+3)} & \longleftarrow & g_{M_2+3} \\
 & & \vdots \\
\vdots & \vdots & g_{M_3} \\
 & & g_{M_3+1} \\
 & & g_{M_3+2} \\
 & & g_{M_3+3} \\
 & & \vdots \\
U_{\gamma(M_4)} & \longleftarrow & g_{M_4} \qquad \mu_1 = 4 \\
 & & g_{M_4+1} \\
 & & \vdots \\
 & & g_{M_5} \\
 & & g_{M_5+1} \\
 & & \vdots
\end{array}
$$

$$\nu_1 = \max\{\gamma(n) : n \leqslant M_{\mu_1}\}$$

Then we define

$$\sigma(\langle \mathscr{F}_0, \mathscr{F}_1 \rangle) = \{U_n : n > \nu_1\} \cap \mathscr{W} \in \mathcal{K}_X.$$

We continue in this way...

Note that, by construction, the ν_k and μ_k are strictly increasing, so the ν_k offer a partition of $\{U_n : n \in \omega\}$ into finite sets.

Since X is k-Menger and we have that this is equivalent to One not having a winning strategy in the k-Menger game on X, the strategy defined above cannot be winning. Hence, Two can make their choices to produce a k-cover and the associated sequences of μ_k and ν_k.

To see that the ν_k establish groupability, let $J \subseteq \omega$ be infinite and $K \in \mathbb{K}(X)$. Then consider $[\mathbf{0}; K, 1/2]$.

We can find $j \in J$, $j > 0$, with

$$[\mathbf{0}; K, 1/2] \cap \{g_\ell : M_{\mu_{j-1}} < \ell \leqslant M_{\mu_j}\} \neq \emptyset.$$

Let $\ell \in (M_{\mu_{j-1}}, M_{\mu_j}]$ be so that $g_\ell \in [\mathbf{0}; K, 1/2]$ and notice that $K \subseteq U_{\gamma(\ell)}$. Since this corresponds to the jth inning, this $U_{\gamma(\ell)}$ is in the jth "block" of the U_n.

Thank You!